

SQL Queries — Part 2: Second Highest Salary

Mastering Nested Subqueries, Aggregations, and Conditional Operators

1. Logic: Finding Second Highest Salary

Suppose we have a set of employee salaries: 10,000 → 20,000 → 30,000 → 40,000 → 50,000. Our objective is to determine the second highest value (40,000).

Step 1: Find Maximum Salary

```
SELECT MAX(Salary) FROM
Employee;
```

Output: 50000

Step 2: Exclude Maximum Salary

```
SELECT Salary FROM Employee
WHERE Salary != ( SELECT MAX(Salary)
FROM Employee );
```

Remaining: 10k, 20k, 30k, 40k

Step 3: Find Maximum of Remaining Salaries

```
SELECT MAX(Salary) FROM Employee WHERE Salary != ( SELECT MAX(Salary) FROM
Employee );
```



Core Logic Rule

Second Highest Salary = Maximum of all salaries excluding the highest salary.

2. Find Employee Name Earning Second Highest Salary

If the employee earning 40,000 is **Nitin**, we query his name by wrapping the second-highest logic inside an outer query filter.

```
SELECT Ename FROM Employee WHERE Salary = ( SELECT MAX(Salary) FROM Employee
WHERE Salary != ( SELECT MAX(Salary) FROM Employee ) );
```

3. Visualizing the Three Query Levels

Level 1

Inner Most: Finds absolute highest salary.

50,000

Level 2

Middle Subquery: Excludes Level 1 and finds next MAX.

40,000

Level 3

Outer Query: Retrieves employee name matching Level 2 result.

Nitin